Naira Currency Classification with CNN

This project focuses on classifying Nigerian Naira banknotes using Convolutional Neural Networks (CNNs). It uses a custom image dataset of Nigerian Naira denominations and implements several CNN architectures including VGG19, ResNet50, and InceptionV3 for comparative performance analysis.

📂 Dataset

The dataset is organized into:

- Train/ — training images
- val/ — validation images
- Test/ — testing images

Each folder contains subfolders representing different currency denominations (₦5, ₦10, ₦20, etc.).

🧠 Models Used

- **Basic CNN**
- **VGG19**
- **ResNet50**
- **InceptionV3**

All models use TensorFlow and Keras for image classification tasks.

🚀 Features

- Image preprocessing and data augmentation
- Fine-tuning pretrained models (include_top=False)
- Training, validation, and test performance evaluation
- Model saving for later reuse

📊 Results

Accuracy and loss curves are plotted for each model, helping compare their performances on the Naira classification task.

🛠 Requirements

- Python 3.8+
- TensorFlow
- Matplotlib
- NumPy